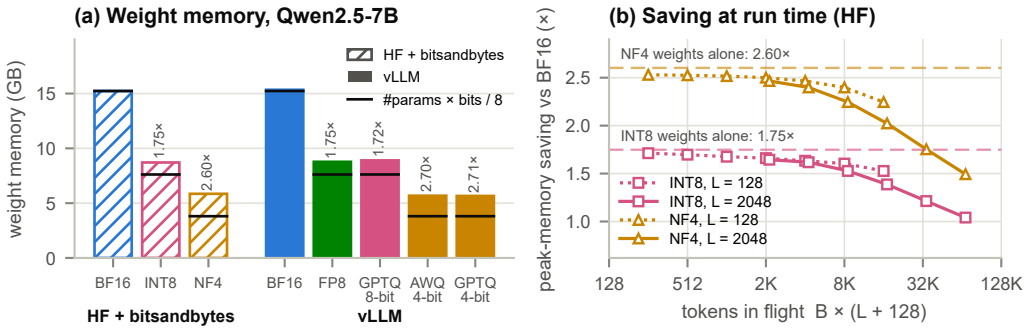

(a) Weight memory, Qwen2.5-7B
HF + bitsandbytes
vLLM
#params × bits / 8
weight memory (GB)
0
5
10
15
1.75×
2.60×
1.75×
1.72×
2.70×
2.71×
BF16
INT8
NF4
BF16
FP8
GPTQ 8-bit
AWQ 4-bit
GPTQ 4-bit
HF + bitsandbytes
vLLM
(b) Saving at run time (HF)
NF4 weights alone: 2.60×
INT8 weights alone: 1.75×
peak-memory saving vs BF16 (×)
1.0
1.5
2.0
2.5
INT8, L = 128
INT8, L = 2048
NF4, L = 128
NF4, L = 2048
128
512
2K
8K
32K
128K
tokens in flight B × (L + 128)